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M 003 THERMAL INSULATION AND LAGGING

1 SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Summary of Works.

All portions of the plant which have an exposed surface and, during normal operation, run at temperatures of 60°C and above, shall be insulated.

The thickness of lagging applied shall be such as to reduce the heat losses to economic levels. Unless otherwise stated the required thickness shall be such that, with a still air temperature of 20°C to 35°C, and if appropriate screened from solar gain, the surface temperature of the lagging insulation material proper before the application of finishing material or paint shall not exceed the temperature of 60°C. Manufacturers standard insulation is acceptable for preengineered packages like GT and ST.

In addition to insulation provided to reduce heat loss to economic levels, insulation, or other measures (such as guards) which the Bidder may propose and the Owner approve, shall be provided for purposes of personnel protection.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”. The codes and standards of the following organizations extracted from Part 4, Subsection 2.5 “Codes and standards” or other international equivalent standards are specifically applicable to the design manufacture and testing of the equipment included in this Specification.

ASTM	American Society of Testing and Materials
ASME	American Society of Mechanical Engineers
NFPA	National Fire Protection Association
UL	Underwriters’ Laboratories, Inc
DIN	Deutches Institut für Normung

3 DESIGN AND OPERATING CONDITIONS

The insulation and lagging shall be designed based on the information stated herein for equipment and for piping on the piping line list (as specified in Part 4, Section 3 Technical Specification, M-001 Piping). The Insulation thickness shall be in accordance with the requirements of Clause 4.2.1.

4 DETAILED REQUIREMENTS

4.1 General

Where insulation is provided for heat conservation it shall reduce the heat loss to an economic minimum. The maximum heat loss shall be 200 W/m² at an ambient temperature acceptable at 30°C and a wind speed of 2 m/s.

4.1.1 Hot Pipe or Equipment Insulation

Insulation applied to all surfaces with a working temperature of 60°C or higher shall be designed to minimize heat losses the outer surface temperature of the insulation shall not exceed 55°C.

4.1.2 Personnel Protection Insulation

For personnel protection all surfaces with a temperature above 50°C and which are within the reach of personnel shall be provided with protection insulation.

4.1.3 Antisweat Insulation

All cold water and gas piping shall be provided with antisweat insulation to prevent condensation on the pipe surfaces.

4.1.4 Conformance

Insulation work, materials and accessories shall conform to the requirements of this Specification.

4.1.5 Methods of Application

During the application of heat insulation the Bidder shall protect adjacent plant, buildings and floors from damage. The plant shall be covered with dust sheets or other protective coverings. The Bidder shall make good any damage at his own expense.

Piping and equipment shall be clean, dry and free from grease, dirt, loose rust or scale before application of insulation materials.

On vertical or near vertical piping all insulation shall be supported in position by means of metal rings, part rings, studs, or cleats at intervals of 950 mm and above each expansion break in the insulation. The insulation supports should not penetrate the insulation more than is absolutely necessary for reliable support purposes.

Preformed materials shall be used for the insulation of pipework but where the pipe size is too large for this to be practicable, suitable mats shall be used.

The insulation blankets shall be stitched in position longitudinal and circumferential by wire of 1mm diameter.

Where flanges and flanged valves are to be insulated the pipework insulation shall be terminated short of the flanges to enable easy withdrawal of the flange bolts without damage to the insulation.

A hole of approximately 7.5 mm diameter shall be formed in the insulation adjacent to flanges extending from the joint to the outside so that any leakage which may occur will be quickly apparent.

Thermocouple pocket bosses, pressure tapings shall be fitted with removable boxes or sections. A vapor barrier in accordance with the recommendations of BS 5970 shall be provided for all surfaces on equipment operating below ambient temperatures.

4.1.6 Materials

All insulating materials shall be of the best quality and suitable for the purpose intended, and shall be of such durability as to be capable of giving effective service for a period of at least thirty years under normal operating conditions.

All insulation for hot surfaces shall be of high thermal efficiency and shall be capable of withstanding the maximum specified working temperatures to which they will be subjected without deterioration and without loss of thermal efficiency.

4.1.6.1 Type of Insulation Materials

The following insulating materials may be used:

- Piping DN $\leq$ 250 preformed shape blankets
- Piping DN $>$ 250 blankets

The mats shall be stable in shape, chemically inert, free of sulfur and alkali, resistant to water and steam, non-flammable and capable of withstanding continuous exposure to the pipe design temperature. The mats used for insulation of stainless steel equipment shall have a chloride content of less than 0.15%. Under no circumstances may asbestos or asbestos containing materials be used.

The material will have the following main characteristics:

Materials shall be free from asbestos and sulphur, chemically neutral, stable at the service temperature and under humidity, non-flammable, imputrescible and resistant to vermin.

For use on stainless steel surface, insulating materials shall be of low halogen content and the chloride shall be less than 6ppm.

In all cases the limit temperature of the insulating materials shall be at least 50 °C higher than the considered fluid temperature.

The choice of the material shall allow :

- Sufficient mechanical strength
- Perfect cut-out of joints
- Exact fit to the shapes of the surfaces to be insulated.

Glasswool insulation shall not be used on surfaces above 350 °C. For surfaces with higher temperatures, this insulating material may be used as a second layer if the temperature of the glasswool will not exceed 350 °C.

Minimal density of insulating materials

- | | | |
|-------------------------------------|---|----------------------|
| - Rockwool | : | 70 kg/m ³ |
| - Glasswool | : | 50 kg/m ³ |
| - Polyurethane and polyisocyanurate | : | 35 kg/m ³ |

- Loose mineral wool after padding : 140 kg/m³

For special purposes such as for turbines, boilers, etc. spray type insulation or insulation brickwork (for example calcium silicate) may also be applied. The special insulation materials shall be stated by the Bidder and their design shall match the overall design and guarantee requirements stated in the specifications.

4.1.6.2 Type of Insulation Setting Materials

The surface cladding shall be made of aluminum sheets specified as “99.5% aluminum grade H24, mill finish surface” manufactured in sheets. The sheet thickness shall be subject to Owner’s approval.

Tank tops shall be provided with insulation strong enough to support a man’s weight.

The sheets shall be secured and connected at the longitudinal seams with at least five stainless steel self-tapping screws per meter run.

Plain sheets for flat surfaces shall not exceed 1 meter square and shall be stiffened by crimping.

At the longitudinal and circumferential joints, the sheets shall overlap by at least 50 mm so as to drain off the liquid and not to trap the liquid in the insulation.

Places at which the metal sheets are penetrated for pipe hangers, thermometer sockets, etc. shall be sealed with funnel-shaped recesses or sheet metal rims.

The seams and penetrations of any sheet metal insulating jacket installed outdoors as well as in the turbine house in areas with risk of water spray etc., shall be sealed against penetration of water by means of a suitable silicon based sealant.

The inside of the aluminum jacket shall be protected against contact corrosion from the wire mesh of the insulating mats by suitable means (e.g. Kraft paper).

In the case of pipe insulation thicknesses greater than 60 mm, where insulation blankets are used spacers shall be provided at maximum intervals of 950 mm to ensure a uniform insulating thickness on all sides and a perfectly circular shape of the sheet metal jacket. The sheet metal jacket shall be supported by support

webs (for pipe diameters below 100 mm) and enclosed support rings (for pipe diameters 100 mm and above). If the enclosed support rings are not provided with ceramic spacers, the spacers made of steel shall be insulated with one heat insulation strip in the case of operating temperatures of up to 200°C, and with two heat insulation strips where the operating temperature exceeds 200°C.

To provide protection against contact corrosion, the external circumference of the support ring shall be fitted with heat insulation tape 1 mm thick with two woven edges.

The use of “Pop” type rivets as a means of securing or supporting insulation, lagging, and/or flashings is not permitted.

Stiffeners, flanges, lugs and other hot projecting parts on insulated surface of all equipment shall be provided with insulation of not less than 38 mm thickness at all points. All doors and frames, and expansion joints etc. shall also be properly insulated to reduce heat radiation.

The insulation and lagging shall be arranged with expansion joints, if required, so that all surfaces will be effectively insulated whether in the hot or cold position and to prevent cracking or distortion of the insulation due to thermal expansion of the equipment.

All support rods piercing through casings shall be insulated in such manner as to be heat leak proof and to minimize heat transfer through the rods. All insulated surfaces of the equipment shall be metal lagged. Metal lagging shall be aluminum, of thickness as specified herein.

4.1.7 Insulation Review

Bidder shall submit to the Owner for review, complete specifications covering materials and methods of application including standard details for approval.

4.2 Piping Insulation

4.2.1 Insulation Thickness

The following table shall be used as a guideline to determine the minimum insulation thickness for all piping and equipment included in the Contract. The

Bidder shall insulate all piping and equipment to obtain a surface temperature of 55°C or less.

AVERAGE THICKNESSES OF INSULATION, IN mm

Pipe Size (mm)	Thermal Insulation Thickness (mm)									
	Medium Temperature °C									
	100	150	200	250	300	350	400	450	500	550
25	30	30	30	40	50	60	70	80	90	100
50	30	30	40	50	60	70	70	90	100	110
80	30	40	50	60	70	80	90	100	110	120
100	40	50	60	70	80	90	100	110	120	130
150	40	60	70	80	90	100	120	130	140	150
200	50	60	80	90	100	120	130	140	150	160
250	50	70	90	100	110	130	140	150	170	180
300	60	80	90	110	120	130	150	160	180	190
350	60	80	100	110	130	140	150	170	180	200
400	60	80	100	120	130	150	160	180	190	210
450	70	90	100	120	140	150	170	180	200	220
500	70	90	110	130	140	160	170	190	210	220
600	70	100	120	130	150	170	190	200	220	240

4.2.2 Insulation Limits

The following insulation Limits for hot piping shall be included in the “Scope Of Works”:

- Drains up to and including last valve, trapped drains up to but not including the trap. This shall include bypass piping up to and including the bypass valve.
- Vent piping up to and including the second valve.

- Relief and safety valve piping up to valve inlet flange.
- All hot piping draining to open funnels up to and including the last valve before funnel or drain pan.
- Test connections to cap.
- Nitrogen connections to and including the first valve from hot source.
- Chemical feed connections back to and including the first valve before the hot header they service.
- Vents and drains up to and including the last redundant valve (not including operating vents).
- Continuous operating vents or drains with temperatures of 55°C and higher to the point of termination.
- Gauge glass connections complete.
- Instrument sensing lines from process connections up to the condensing chambers.
- All projecting hanger lugs and hanger clamps on piping, with operating temperatures of 150°C and above, and on all outdoor heat insulated piping, shall be completely insulated with the same type and thickness of insulation as that specified for the pipeline in which the hanger is located.

4.3 Equipment Insulation

4.3.1 Insulation of Flanges, Valves and Fittings

All flanges, valves and fittings shall be provided with two-piece or multi-part caps made of aluminum sheet of the specified thickness. Each piece or part shall be double jacketed and the various parts shall be held together by quick release clamps or lever hooks to facilitate assembly and disassembly.

All caps of the welded-in fittings shall be made longer by approximately twice the insulation thickness so that the welding seams will be exposed after removal of the cap.

All manholes shall be provided with heavy duty hinged insulated covers wherever possible. Such covers shall be secured with easily accessible clamps.

4.3.2 Insulation of Tanks and Process Equipment

Tanks and process equipment shall be insulated in the same way as pipes, except that the insulating material shall not be attached by wire but by using strong galvanized steel bands.

Spacers shall be welded to process equipment only if this is essential for satisfactory retention of the insulation. Welding of spacers to process equipment is subject to approval by its manufacturer in writing.

4.3.3 Insulation for Personnel Protection

Wherever insulation is necessary only for the protection of personnel, it shall be applied around that portion of the pipeline length or to that surface of the equipment that is located within 2.50 m above the passage way floor, or within 1.20 m horizontally to the side or at the end of any floor, platform, walkway, stair or ladder.

Where necessary, drain lines and valves shall be provided with a contact guard of minimum 30 mm thickness against accidental contact, and this shall be installed in the same way as the other insulation.

Iron and steel steam trap bodies shall not be insulated, but shall be provided with personnel protection shields. The shield shall be fabricated from minimum 1.3 mm thick expanded galvanized steel.

All sharp edges must be folded back on the material. Shields shall be of removable type, employing screws and nuts for servicing of the steam trap. Bidder shall submit typical sketches to the Owner for review prior to installation

4.3.4 Prefabricated Insulation Panel Construction

Prefabricated insulation panels shall be of sandwich construction consisting of an outer lagging of ribbed aluminum (which will be exposed to the elements) followed by a layer of mineral wool insulation block, aluminum foil, and wire mesh.

The panel assembly shall be held together by insulation holding devices which shall be welded to the inside of the exterior lagging. The insulation, foil and wire mesh shall be impaled over insulation holding devices in the order

indicated, after which a speed washer shall be applied. Insulation holding devices must be 25 mm longer than the specified thickness of insulation. This excess length must be bent over after the installation of the washers. There shall be no free play of the sandwiched materials after the application of the washers.

The interior side of the panels (hot side) shall have all insulation completely covered with aluminum foil and wire mesh.

Panels shall be constructed in such a manner that there will be a tight butt of all the insulation blocks after fabrication and installation.

4.3.4.1 Materials

Outer lagging shall be 1.00 mm thick, 100mm center-to-center, 25mm depth ribbed aluminum. All flat aluminum and aluminum flashing, where required, shall be 1.3 mm thick. Aluminum lagging shall be 3004 alloy with 7072 alloy cladding. Core materials shall conform to the latest edition of ASTM Specification B209. The finish of the aluminum lagging shall be stucco embossed.

Aluminum foil shall be not less than 0.06 mm thick.

4.3.4.2 Wire Mesh

- Mesh for panels shall be hot dip galvanized steel wire.
- Galvanizing shall be performed after fabrication of the mesh.

4.3.5 Prefabricated Thermal Acoustical Panel Constructions

Prefabricated thermal-acoustic insulation panel shall be of sandwiched construction consisting of an outer lagging of 1.00 mm thick ribbed, stucco embossed aluminum (which will be exposed to the elements), followed by a layer of 25 mm thick mineral wool insulation block, a layer of lead sheet (4.9 kg/m²) and a second layer of 25 mm thick mineral wool insulation block, 0.06 mm thick aluminum foils and a layer of 16 gauge, 51 mm x 66.7 mm rectangular, hot dipped, galvanized, wire mesh. The total thickness of the panel will be 51 mm plus the outer lagging.

Outer lagging shall be 1.00mm thick, 102 mm center-to-center, 25 mm depth, ribbed aluminum. Flashing where required in panel work shall be 1.3 mm

thick aluminum. Aluminum lagging shall be 3004 alloy with 7072 alloy claddings. Core material shall conform to the latest edition of ASTM Specification 8209.

Insulation holding devices shall be not less than 2.8 mm thick. Insulation holding devices shall be placed on not greater than 450 mm centers both ways.

Washers for retention of sandwiched material shall be 76 mm round or square and of a material that is compatible with the mesh.

4.4 Insulation Fastenings

All surfaces to be insulated, except piping and tubes, shall have clips, welding studs, pins or other suitable means spaced on centers as hereinafter specified for attachment of insulation holding wire for impaling mineral wool blocks or blankets. All materials required for proper support and fastening of insulation to all surfaces shall be provided.

Mineral wool block, board or blankets shall be applied by impaling on self-welded studs or pins and retaining the insulation with speed washers. Studs or pins shall be in a set pattern with spacing not in excess of 450 mm centers in both directions on the top and sides, and 300 mm centers in both directions on bottom of the insulated surfaces. The pin length shall be 25 mm greater than the thickness of insulation. After the insulation and speed washers are in place the pins shall be bent over and not cut.

Calcium silicate block insulation shall be securely wired in place. Welding studs or drilled clips shall be welded to the surfaces in a set pattern with spacing not in excess of 450 mm centers in both directions on top and sides and 300 mm centers in both directions on bottom of insulated surfaces for attaching insulation tie lacing wires.

Joints in insulation shall be tightly butted so as to produce a monolithic continuous insulation. Edges of metal mesh covered blankets shall be tied together with wire.

4.5 Lagging

All insulated surfaces of straight pipe, fittings, valves, flanges, equipment etc. shall be lagged with half-hard Aluminum Association Sheet No.3003, with a minimum thickness of 0.4 mm, stucco embossed, with a factory attached

moisture barrier of 18 kg kraft paper coated with 25.4 micron thickness of polyethylene. The use of asphaltic type moisture barrier will not be permitted. The above shall not apply to cold equipment.

Insulation with factory attached metal lagging will not be acceptable except where specified otherwise.

Metal lagging shall be properly designed and installed so as to completely cover all insulation and to provide for expansion and contraction so the lagging shall lay flat and straight with the equipment hot or cold.

For outdoor applications, the metal lagging shall be completely waterproof with proper flashings installed so that water cannot reach the insulation. The lagging shall be designed and installed to provide for complete water runoff with no pockets where water can collect.

Removable panels of doors shall be provided in the metal lagging to provide for removal of handhold closure fittings, manholes etc.

The method of fastening metal lagging shall be acceptable to the Owner. The spacing of fasteners shall be in a set pattern both vertically and horizontally and shall be such as to prevent ratting of the lagging due of effect of equipment or piping vibration.

Dissimilar metal that may cause galvanic corrosion shall be properly isolated.

Metal lagging fastening screws shall be cadmium-plated or stainless steel for the aluminum, aluminized sheeting or galvanized sheeting. Nylon, neoprene or vinyl washers shall be-used. All fastening screws shall be of the type that will not stain the lagging. Full-threaded stud setlock fastener and blind rivets are not acceptable methods of fastening lagging.

The alloy temper shall be such that the material can be bent flat on itself without cracking.

4.6 Insulation Samples

A sample of each type of pipe and equipment insulation, together with the manufacturer documentation and thermal efficiency tables or curves based on a recognized authority, shall be submitted to the Owner for information prior to installation.

4.7 Ventilation and Air Conditioning Insulation

All air conditioning ductwork shall be insulated as specified herein.

4.7.1 Internal Duct Liner

Insulation shall be 25 mm thick, 48 kg/m³ minimum density fiberglass board impregnated with NFPA approved fire retardant, vapor proof coating having a smooth surface on the side exposed to the air-stream.

Adhesive shall be vapor barrier mastic and shall be compatible with the above insulation material as recommended by the insulation manufacturer.

4.7.2 Indoor External Insulation for Rectangular Ducts

Insulation shall be not less than 25 mm thick, and not less than 64 kg/m³ density fiberglass board with factory applied laminate of presized, finelywover glass cloth and foil1 having a white finish suitable for painting.

Sealing tape shall not be less than 76 mm wide and shall be the insulation manufacturer's recommended vapor proof pressure sealing tape. Color shall match the insulation facing.

Adhesive shall be white, vapor barrier mastic and shall be compatible with the above insulation materials as recommended by the insulation manufacturer.

The entire insulation system shall meet NFPA 90A Fire Hazard Classification.

4.7.3 Indoor External Duct Insulation for Round Ducts

Insulation shall be 25 mm thick, preformed fiberglass, pipe insulation with factory applied laminate of presized, finely woven, glass cloth and foil, having a white surface suitable for painting. Fiberglass insulation shall have a density of not less than 64 kg/m³.

Sealing tape shall be not less than 76 mm wide and shall be the insulation manufacturer's recommended vapor proof pressure sealing tape. Color shall match the insulation facing.

Adhesive shall be white, vapor barrier mastic and shall be compatible with the above insulation materials as recommended by insulation manufacturer.

The entire insulation system shall meet NFPA 90A Fire Hazard Classification.

4.7.4 Outdoor External Duct Insulation and Finish

Insulation shall be 25 mm foam plastic sheet insulation finished with two (2) coats of manufacturer's recommended finish for outdoor service.

5 GUARANTEES

The Bidder shall guarantee that the types, thickness, combinations and methods of application of the insulating materials called for in this Specification are suitable for the specified service conditions. Therefore the Bidder shall warrant that the materials furnished by him are suitable for their specified service and will maintain their thermal efficiency, physical qualities and mechanical strength without deterioration under the conditions for which they will be used.

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